Different forms of symbiotic integration

Ruthia magnifica

Clam endosymbiont;
core metabolic pathways encoded

Strain Cm chromosome
1.16 Mb

Elysia chlorotica

Plastids from V. litorea;
acquired anew by feeding

No inherited symbiont
chromosome represented

Carsonella ruddii

Obligate psyllid symbiont;
strong genome reduction

Strain PV chromosome
160 kb

Human mitochondrion

Organelle; depends on
nuclear-encoded proteins

Human reference mtDNA
16.6 kb

Different associations, not evolutionary stages